Sales Performance Analysis

Superstore Retail Data: Complete Project Report

Data Analysis Capstone Project

Karthik CG
3rd BCA, Sri JCBM College, Sringeri
GitHub: github.com/karthik2252/sales-analysis-capstone

Contents
1. Introduction and objectives
2. Tools and technologies
3. Dataset description
4. Data preparation
5. Exploratory data analysis
6. Data visualization
7. Dashboard
8. Business insights
9. Recommendations
10. Git and GitHub workflow
11. Limitations, conclusion and future scope

Abstract

This project analyses the Superstore retail dataset of 9,994 order lines (2014 to 2017) to find out what drives and what hurts profit. The business earned total sales of 2,297,201 and total profit of 286,397, a profit margin of 12.47%. The analysis covers data inspection and preparation, descriptive statistics, correlation, outlier detection, six visualizations and an interactive dashboard. The main finding is that profit is hurt far more by heavy discounting and a few loss-making sub-categories than by weak sales volume. Five data-driven recommendations are given.

1. Introduction and Objectives

Retail businesses sell thousands of products across regions, segments and seasons, and growth in sales does not always bring growth in profit. This project studies one retail dataset end to end, from raw file to business decisions, using the tools and workflow taught during the Data Analysis internship.

Problem statement

Sales are growing, but the profit margin is only 12.5%. Which categories, regions, products and discount levels drive or hurt profit, and what should the business change?

Objectives

- Select and understand a real-world retail dataset.
- Clean and prepare the data for analysis.
- Perform exploratory data analysis with statistics, correlation and outlier detection.
- Create at least five meaningful visualizations.
- Build an interactive dashboard with KPI cards, charts and slicers.
- Identify at least five business insights and two recommendations.
- Document the project on GitHub using a proper commit workflow.

2. Tools and Technologies

Tool	Used for
Python (Pandas, NumPy)	Data loading, cleaning, statistics
Matplotlib and Seaborn	Charts and heatmap
Power BI Desktop	Interactive dashboard (SuperSt.pbix)
HTML, JavaScript	Browser version of the dashboard (dashboard.html)
Git and GitHub	Version control and project hosting
Kaggle	Source of the dataset

3. Dataset Description

The Superstore dataset from Kaggle contains 9,994 rows and 21 original columns. Each row is one product line within a customer order. The data covers 5,009 orders from 793 customers in the United States between January 2014 and December 2017, in 4 regions, 3 categories and 17 sub-categories.

Column	Type	Meaning
Row ID	Integer	Row number
Order ID, Customer ID, Product ID	Text	Identifiers
Order Date, Ship Date	Date	When the order was placed and shipped (stored as text, converted)
Ship Mode	Text	Standard, Second, First Class or Same Day
Customer Name, Segment	Text	Customer and segment (Consumer, Corporate, Home Office)
Country, City, State, Postal Code, Region	Text / Integer	Location of the order
Category, Sub-Category, Product Name	Text	Product hierarchy
Sales	Decimal	Revenue of the order line
Quantity	Integer	Units sold
Discount	Decimal	Discount given (0 to 0.8)
Profit	Decimal	Profit or loss of the order line

4. Data Preparation

Step	Result
Inspection	Shape (9,994, 21); head, data types and summary checked
Missing values	0 missing values in all 21 columns. The code fills text columns with the mode and numeric columns with the median if any appear
Duplicates	0 duplicate rows. The code drops them if present
Incorrect records	Checked Sales > 0, Quantity > 0 and Ship Date on or after Order Date; no rows needed removal
Data types	Order Date and Ship Date converted from text to datetime; Postal Code treated as text
New columns	Year, Month, Month Name, Profit Margin % (25 columns in the cleaned file)
Output	data/superstore_cleaned.csv

```
df = pd.read_csv("data/superstore.csv", encoding="latin1")
df.info(); df.isnull().sum(); df = df.drop_duplicates()
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Profit Margin %"] = df["Profit"] / df["Sales"] * 100
```

5. Exploratory Data Analysis

Descriptive statistics

Statistic	Sales	Quantity	Discount	Profit
Mean	229.86	3.79	0.16	28.66
Median	54.49	3.00	0.20	8.67
Std dev	623.25	2.23	0.21	234.26
Min	0.44	1.00	0.00	-6,599.98
Max	22,638.48	14.00	0.80	8,399.98

Correlation analysis

Correlation	Sales	Quantity	Discount	Profit
Sales	1.00	0.20	-0.03	0.48
Quantity	0.20	1.00	0.01	0.07
Discount	-0.03	0.01	1.00	-0.22
Profit	0.48	0.07	-0.22	1.00

Sales and Profit are moderately positively correlated (0.48), so bigger sales help but do not guarantee profit. Discount and Profit are negatively correlated (-0.22), the clearest warning in the data. Quantity has almost no relationship with profit (0.07).

Outliers

Using the IQR rule on Profit, 1,881 order lines (18.8%) are outliers. The largest loss is -6,600 and the largest profit is 8,400. These are genuine large orders, so they were kept in the analysis.

Performance by year

Year	Sales	Profit	Margin %
2014	484,247	49,544	10.2
2015	470,533	61,619	13.1
2016	609,206	81,795	13.4
2017	733,215	93,439	12.7

Performance by category

Category	Sales	Profit	Margin %
Technology	836,154	145,455	17.4
Office Supplies	719,047	122,491	17.0
Furniture	742,000	18,451	2.5

Performance by region

Region	Sales	Profit	Margin %
West	725,458	108,418	14.9
East	678,781	91,523	13.5
South	391,722	46,749	11.9
Central	501,240	39,706	7.9

Performance by segment

Segment	Sales	Profit	Share of sales %
Consumer	1,161,401	134,119	50.6
Corporate	706,146	91,979	30.7
Home Office	429,653	60,299	18.7

Profit by sub-category

Sub-category	Sales	Profit
Copiers	149,528	55,618
Phones	330,007	44,516
Accessories	167,380	41,937
Paper	78,479	34,054
Binders	203,413	30,222
Chairs	328,449	26,590
Storage	223,844	21,279
Appliances	107,532	18,138
Furnishings	91,705	13,059
Envelopes	16,476	6,964
Art	27,119	6,528
Labels	12,486	5,546
Machines	189,239	3,385
Fasteners	3,024	950
Supplies	46,674	-1,189
Bookcases	114,880	-3,473
Tables	206,966	-17,725

Profit by discount band

Discount	Order lines	Avg profit per line	Total profit
0%	4,798.0	66.9	320,988
1-20%	3,803.0	26.5	100,785
21-40%	460.0	-77.9	-35,817
Above 40%	933.0	-106.7	-99,559

Orders with a discount of 30% or more number 1,393; 96.8% of them lose money, a combined profit of -135,376. This is the single biggest drain on profit.

6. Data Visualization

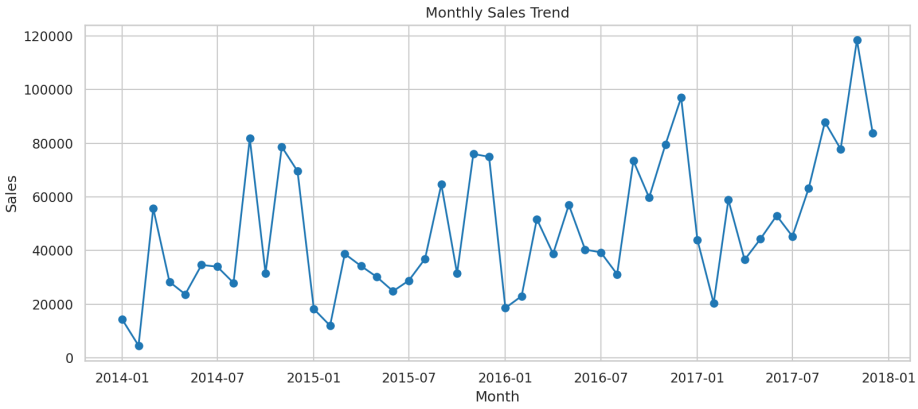


Chart 1: Monthly sales trend. Sales rise every November and December.

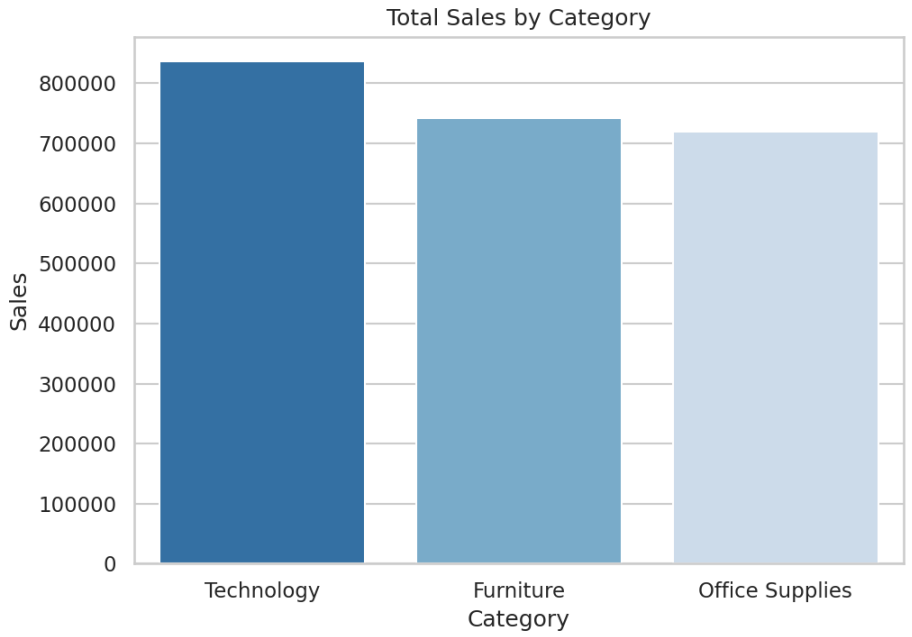


Chart 2: Total sales by category. Technology is highest.



Chart 3: Total profit by region. West is highest, Central lowest.

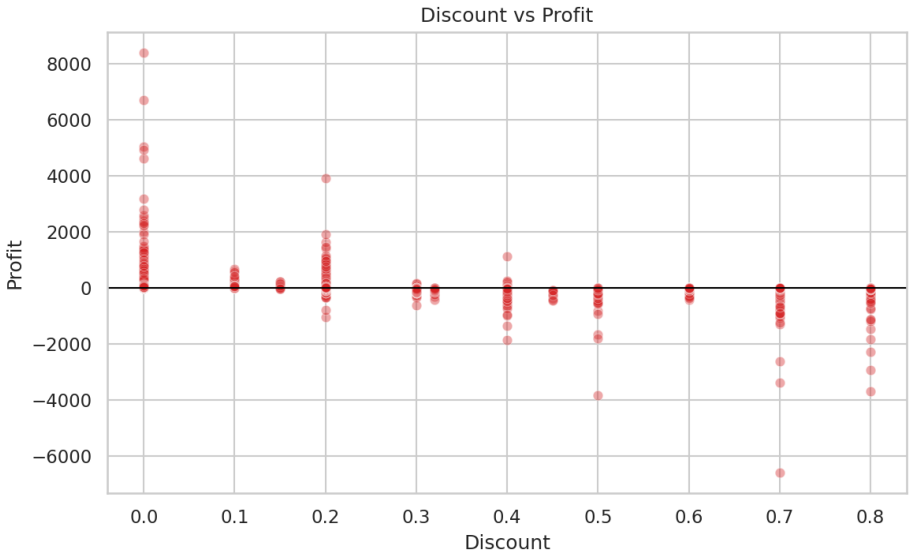


Chart 4: Discount vs profit. Orders above about 30% discount are mostly losses.

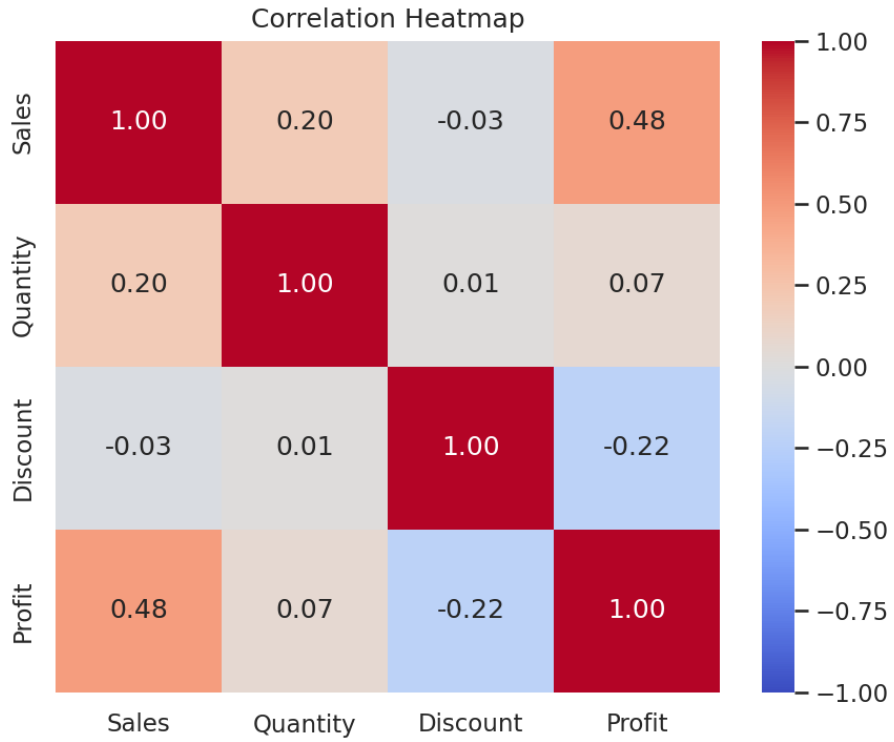


Chart 5: Correlation heatmap of Sales, Quantity, Discount and Profit.

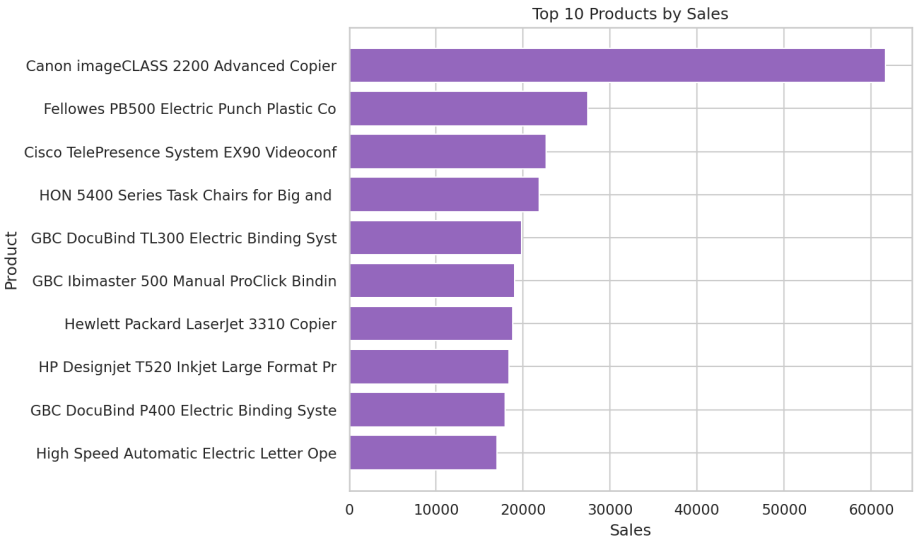
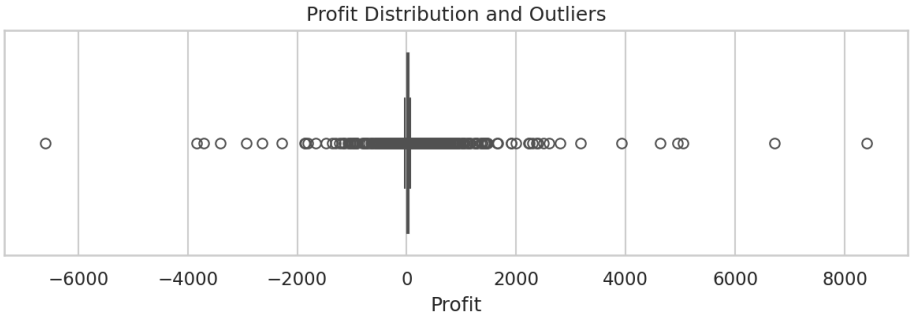


Chart 6: Top 10 products by sales.



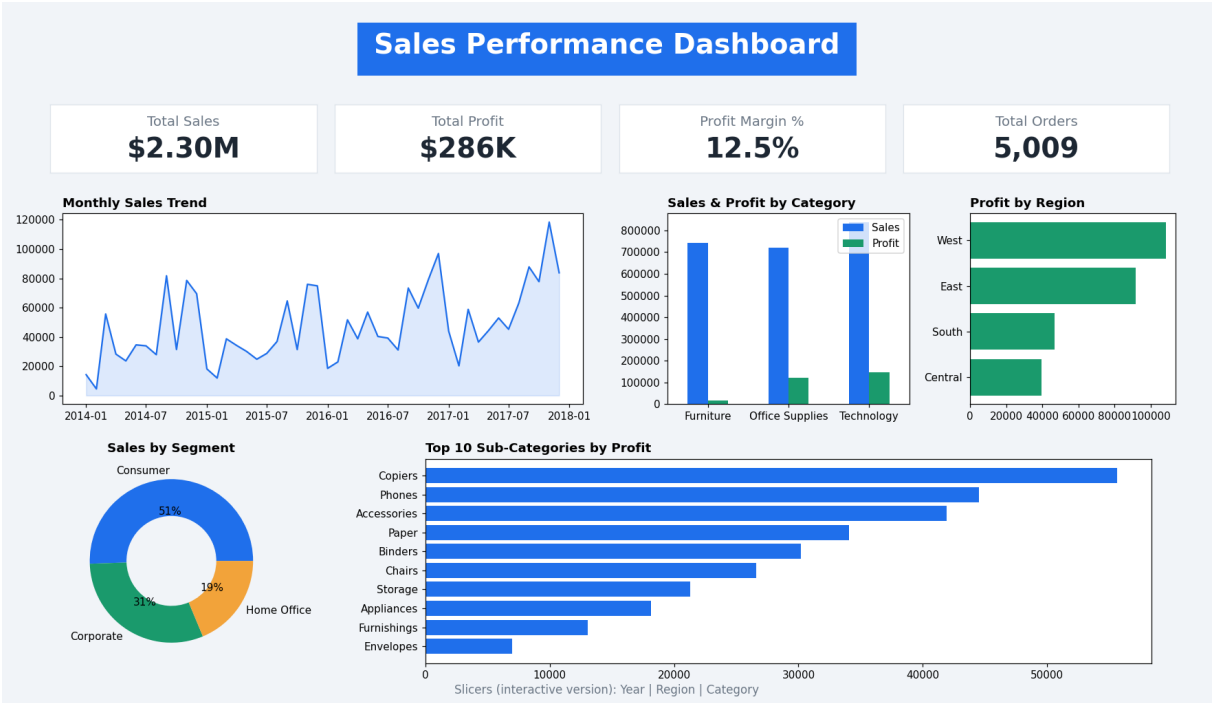
Bonus: Profit distribution and outliers.

7. Dashboard

The Power BI dashboard (SuperSt.pbix, titled Super Store Dashboard: Executive Overview) is a single 1280 by 720 page. It contains five KPI cards (Revenue, Profit, Profit Margin, Quantity, Average Discount) with comparison cards against the previous year and month, seven visuals and slicers:

Visual	Shows
Area chart	Revenue over time (date hierarchy)
Donut chart	Revenue by segment
Scatter chart	Discount against quantity by product
Column chart	Revenue by month
Bar chart	Revenue by category, sub-category and product
Map	Revenue by state
Table	Revenue, profit, quantity, margin and discount by product
Slicers	Region, and a date slicer (Year, Quarter, Month, Day)

A browser version (dashboard.html in the repository) shows 4 KPI cards, 5 charts and Year, Region and Category slicers using the same data. The picture below is a summary overview of the main KPIs and charts made from the dataset.



Dashboard overview (summary picture of the main KPIs and charts)

8. Business Insights

- **Seasonality:** sales peak in November (352,461) and December (325,294) and are lowest in February (59,751).
- **Category gap:** Technology earns 145,455 profit, but Furniture earns only 18,451 on 742,000 sales (about 2.5% margin).

- **Loss-making sub-categories:** Tables (-17,725), Bookcases (-3,473) and Supplies (-1,189). Copiers (55,618), Phones (44,516) and Accessories (41,937) earn the most.
- **Discounts destroy profit:** 1,393 orders with 30%+ discount lose money 96.8% of the time, a combined -135,376.
- **Regional gap:** West earns 108,418 (margin 14.9%), Central only 39,706 (margin 7.9%).
- **Customer concentration:** the top 20% of customers generate 48.0% of all sales.
- **Growth:** sales rose 51% from 484,247 in 2014 to 733,215 in 2017, and profit from 49,544 to 93,439.

High and low performers

High: Technology, West and East regions, Copiers, Phones, Accessories, Consumer segment. Low: Tables, Bookcases, Supplies, Central region, and every order with a 30% or higher discount.

9. Recommendations

- **Cap discounts at about 20%** and require manager approval above 30%. Losses from 30%+ discount orders are nearly half of total profit.
- **Reprice or drop** Tables, Bookcases and Supplies by raising prices, renegotiating supplier cost or reducing stock.
- **Plan seasonal campaigns:** start festive promotions early and stock up before November; use light promotions in January and February.
- **Improve the Central region** by reviewing its discount levels and product mix against West.
- **Loyalty programme** for top customers and a push on high-margin Technology products.

10. Git and GitHub Workflow

The project repository is github.com/karthik2252/sales-analysis-capstone. Work was committed in four meaningful steps using the clone, add, commit, push and pull workflow:

#	Commit message	Files
1	Add raw dataset	data/superstore.csv
2	Add data cleaning and EDA script	analysis.py, data/computed_insights.txt
3	Add visualizations and dashboard	images/, dashboard.html, Power BI file
4	Add insights, recommendations and final documentation	README.md, docs/

11. Limitations, Conclusion and Future Scope

Limitations

- The data covers four years of one US retailer, so results may not generalise.
- Profit is analysed per order line; shipping and overhead costs are not in the data.
- The analysis shows association, not proof of cause (for example, discounts may be given on already low-margin items).

Conclusion

Profit depends more on discount discipline and product mix than on sales volume. Sales grew 51% in four years, yet the margin stays near 12.5% because heavy discounts and three loss-making sub-categories absorb a large share of earnings. Fixing these two areas could raise profit without needing any extra sales.

Future scope

- Forecast monthly sales with a time-series model.
- Segment customers by value and buying behaviour.
- Test the effect of a discount cap with a what-if analysis in Power BI.

Author: Karthik CG | 3rd BCA, Sri JCBM College, Sringeri | karthikankush59@gmail.com | github.com/karthik2252